Additive Observable Contrast: Maximum-Difference Witnesses for Streaming Physical States

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***Abstract*—Principal components maximize variance, although the direction of largest variance need not distinguish conditions. We formulate the original maximum-class-difference motivation of Eigen-Component Analysis as *additive observable contrast* (AOC). Each sample is encoded as a positive trace-one operator; per-condition sums and masses are exactly mergeable; and the bounded effect with the largest expectation change is the positive spectral projector of the state difference. The resulting scalar is trace distance, rank budgets have an analytic Ky Fan solution, and the multiclass extension is the minimum-error positive-operator-valued-measurement semidefinite program. A predictable online witness can be coupled to a betting martingale when the reference state is known. Numerical audits match batch, merged, kernel, and semidefinite formulations to at most 6.7×10^{-9}. Matched physical tests expose the intended regime rather than universal dominance: under exact sign symmetry, a linear Ising classifier remains at 50% while AOCobtains 99.98% and recovers the magnetization mode; a rank-one structural witness obtains 0.977 ROC AUC and 0.993 overlap with the analytical damage mode; and an optical analyzer raises state-discrimination success from 50% to 95%, reproduced by finite-shot Qiskit Aer simulation. The method is therefore best viewed as a mergeable, interpretable measurement-design primitive for state discrimination, change detection, and mode localization, not as a universal classifier.**

***Index Terms*—change detection, density operators, interpretable learning, online algorithms, quantum-inspired computing, state discrimination**

I. Introduction

VARIANCE is valuable when the dominant fluctuation is also the phenomenon of interest. It is misleading when a nuisance direction has large variance, or when two conditions have the same mean and differ only in power or coherence. The original Eigen-Component Analysis (ECA) work sought features with large class difference rather than large pooled variance [1]. The question left open is what "difference" means, which constraints make it operational, and how the object should be updated as observations arrive one at a time.

We answer these questions using positive operators. A normalized outer product is simultaneously a second-order feature state, a kernel mean embedding, an optical coherency state, and—when the data are quantum—a density operator. A bounded quadratic readout is an *effect* $0 \preceq E \preceq I$. The optimal difference is then not an analogy to quantum theory; it is the binary state-discrimination problem solved by Helstrom [2], [3]. This identification prevents an incorrect novelty claim while supplying a precise foundation.

The contribution is a synthesis oriented toward streaming physical data:

1) We express maximum-difference feature learning as the positive part of an additive state contrast, including exact rank-constrained and multiclass forms.
2) We expose exact batch–online–merge equivalence and distinguish a valid predictable sequential witness from a same-sample adaptive score.
3) We validate algebra, mode recovery, finite-shot measurement, and false alarms from stored raw records, with explicit claim boundaries.

This positioning addresses a practical distinction from covariance generalized eigenmethods such as common spatial patterns (CSP) [4]: CSP optimizes a variance ratio under a reference metric, whereas AOC optimizes an absolute bounded expectation gap after trace normalization.

II. Positive-State Formulation

A. Additive encoding

For a nonzero feature vector $\phi(x) \in \mathbb{C}^d$, define

$$R(x) = \frac{\phi(x)\phi(x)^\dagger}{\|\phi(x)\|_2^2}, \quad R(x) \succeq 0, \quad \operatorname{Tr} R(x) = 1. \tag{1}$$

Condition c is represented by

$$S_c = \sum_i w_i R(x_i), \quad m_c = \sum_i w_i, \quad \rho_c = S_c/m_c. \tag{2}$$

The statistic (S_c, m_c) is exactly additive. It is constant in the number of samples, but not private: outer-product sums can leak information, and privacy requires a separate threat model and mechanism.

B. Binary optimal effect

Let $\Delta = \rho_1 - \rho_0$ and let $\Delta = \Delta_+ - \Delta_-$ be its Jordan decomposition.

Theorem 1 (maximum bounded contrast). For Hermitian Δ,

$$\max_{0 \preceq E \preceq I} \operatorname{Tr}(E\Delta) = \operatorname{Tr}(\Delta_+), \qquad E^\star = \mathbf{1}_{\Delta>0}. \tag{3}$$

Proof: Because $0 \preceq E \preceq I$, $\operatorname{Tr}(E\Delta_+) \leq \operatorname{Tr}\Delta_+$ and $-\operatorname{Tr}(E\Delta_-) \leq 0$. Equality is obtained by the support projector of Δ_+, which is orthogonal to the support of Δ_-. ■

For trace-one inputs, $\operatorname{Tr}\Delta = 0$ and therefore

$$\operatorname{Tr}\Delta_+ = \tfrac{1}{2}\|\Delta\|_1. \tag{4}$$

With equal priors this gives $P^\star_{\text{succ}} = 1/2 + \|\Delta\|_1/4$, the Helstrom success probability [2]. Quantum Chernoff theory addresses the distinct many-copy error exponent [5].

Working paper. Code, fixed-seed records, and generated figures are available in the accompanying public repository. This manuscript does not claim a hardware implementation or a privacy guarantee.

Corollary 1 (rank budget). If $\lambda_1 \geq \cdots \geq \lambda_d$ are the eigenvalues of Δ, then

$$\max_{\substack{0 \preceq E \preceq I \\ \mathrm{rank}(E) \leq r}} \mathrm{Tr}(E\Delta) = \sum_{j=1}^{r} \max(\lambda_j, 0). \tag{5}$$

The effect projects onto the retained positive eigenvectors. This is a measurement-capacity constraint, not only a statistical penalty.

C. Multiclass observations

Independent one-versus-rest effects need not sum to a realizable measurement. For K class states and priors π_c, the coherent extension is

$$\begin{aligned} \max_{\{E_c\}} \quad & \sum_{c=1}^{K} \pi_c \,\mathrm{Tr}(E_c \rho_c), \\ \text{s.t.} \quad & E_c \succeq 0, \qquad \sum_c E_c = I. \end{aligned} \tag{6}$$

This is the established minimum-error state-discrimination SDP [3]. Its no-information baseline is $\max_c \pi_c$. Our implementation returns primal feasibility diagnostics; tests obtain unit success on orthogonal states, the prior baseline on identical states, and the Helstrom value for $K = 2$.

D. Kernel interpretation

For normalized feature rows, the projective kernel

$$k(x, z) = |\langle \phi(x), \phi(z) \rangle|^2 \tag{7}$$

has mean embedding ρ. Consequently,

$$\mathrm{MMD}_k^2(P, Q) = \|\rho_P - \rho_Q\|_F^2. \tag{8}$$

MMD [6] and AOC use the same contrast operator but different norms: MMD uses Hilbert–Schmidt magnitude, while AOC uses trace magnitude and produces a bounded witness. Neither norm guarantees universally better finite-sample power.

III. Streaming and Sequential Detection

A. Exact state operations

An arrival performs $(S, m) \leftarrow (S + wR, w + m)$. Disjoint summaries merge by addition. A retained observation can be removed, a fixed window can add and subtract, and forgetting applies $(S, m) \leftarrow (\gamma S, \gamma m)$. Associativity implies that centralized batch, arbitrary partitions, and arbitrary merge trees return the same empirical state up to floating-point roundoff. Communication of a dense Hermitian state costs $O(d^2)$ numbers per merge, independent of sample count; diagonal, sparse, or low-rank structure can reduce this cost.

B. Predictable learned witness

Same-sample optimization invalidates ordinary null calibration. Let E_t be measurable with respect to observations before time t, and suppose the reference state ρ_0 is known. The score

$$z_t = \mathrm{Tr}[E_t(R_t - \rho_0)] \in [-1, 1] \tag{9}$$

has conditional mean zero under an independent stationary null. For $0 < \lambda < 1$,

$$M_t(\lambda) = \prod_{i=1}^{t} (1 + \lambda z_i) \tag{10}$$

is a nonnegative martingale. Mixtures over λ and candidate start times remain e-processes, yielding anytime evidence under their stated assumptions [7]. A plug-in reference estimate is diagnostic unless its uncertainty is incorporated. Recent work already develops universal quantum-observable change detection with classical shadows and e-detectors [8], [9]; therefore generic online quantum change detection is not claimed as novel here.

C. Complexity and stability

Dense accumulation is $O(d^2)$ per state and a complete eigendecomposition is $O(d^3)$. A rank-r Lanczos update can use matrix–vector products without a full decomposition. Witness orientation is stable only when the positive spectral subspace has an eigengap; Davis–Kahan perturbation theory bounds its rotation by estimation error divided by that gap [10]. A small gap should therefore be reported as an identifiability warning even when the scalar trace distance is stable.

IV. Reproducible Physical Tests

A. Protocols

All scripts use fixed seeds, write raw CSV files, save software and Git metadata, and hash outputs. The algebra audit samples random density operators, random arrival orders, and random merge trees. The sequential audit uses a known reference, previsible rank-one witnesses, and an alarm threshold $1/\alpha$ with $\alpha = 0.01$.

The Ising test uses 12×12 Wolff samples at $T = 1.6$ and 3.5. Every configuration is paired with its global negative, making the empirical mean exactly zero in both classes. The structural test is a 12-mass chain under zero-mean thermal excitation; one spring loses stiffness. The optical test uses diagonal/antidiagonal polarization states with visibility 0.9 and $16\,384$ shots per state in Qiskit Aer [11]. These are controlled physical models, not field deployments.

B. Results and interpretation

The algebra audit finds maximum batch/additive density error 3.60×10^{-16}, merge-mass error 9.95×10^{-14}, analytic/SDP Jordan gap error 6.67×10^{-9}, trace identity error 2.22×10^{-16}, and kernel/operator error 2.78×10^{-16}.

Figure 1 and Table I show the intended mean-blind regime. AOC reaches 0.9998 ± 0.0007 accuracy and its leading vector has mean squared overlap 0.9991 with the uniform magnetization mode. The comparison is not universal dominance: the

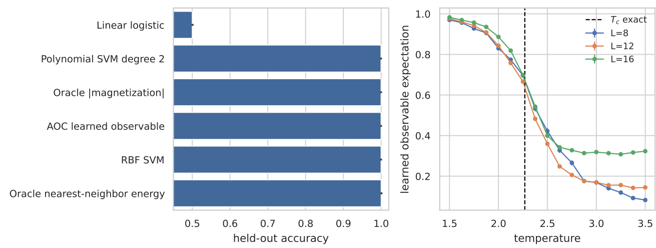

Fig. 1. Left: matched Ising classification. Right: the learned observable tracks the finite-lattice order response. Linear failure is forced by exact Z_2 pairing; nonlinear baselines correctly solve the task.

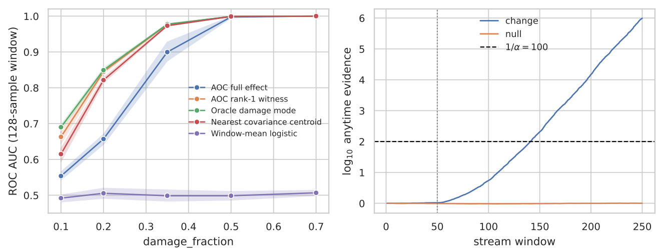

Fig. 2. Structural damage under zero-mean excitation. A rank-one capacity constraint approaches the planted mode and improves over the unrestricted positive subspace.

RBF SVM ties AOC, the degree-two SVM reaches 0.9996, and the physical energy oracle is perfect. The contribution is analytic recovery of an interpretable order mode from a mergeable state.

At 35% spring damage and 128-sample windows, Fig. 2 gives ROC AUC 0.9768 for rank-one AOC, 0.9733 for nearest covariance centroid, 0.9774 for the oracle damage mode, and 0.4985 for window-mean logistic. The learned/analytical damage-mode overlap is 0.9935 on average. Full positive-subspace AOC is worse (0.8996 AUC), demonstrating that maximizing population mean gap need not minimize finite-window decision error when extra modes add variance.

In Fig. 3, the learned effect has Bloch vector $(1, 0, 0)$. The trace distance is 0.9, giving theoretical success 0.95 rather than 0.5 for fixed horizontal/vertical readout. Across 20 seeds, Aer gives 0.950009 ± 0.000996 with maximum probability error 0.003125, one qubit, and transpiled depth two. This is a measurement-basis result, not a quantum speedup claim.

In the general sequential audit, 2 of 500 stationary streams alarm (empirical rate 0.004; exact binomial 95% upper bound 0.0144), while all 200 changed streams alarm with median delay 30. In the more conservative structural stream, 0 of 120 nulls and all 120 changes alarm, with median delay 89 after a 70% damage change. These finite simulations audit the implementation but do not extend the theorem beyond a known independent reference.

V. Scope, Limitations, and Applications

The outer-product encoding loses global sign and phase. This is beneficial in the Ising, power, and coherency tasks but makes phase-coded or homometric signals indistinguishable. Higher tensor powers or phase-retaining feature maps are needed. Dense storage scales as d^2, eigensolvers can dominate, and empirical states can be poorly conditioned when d is large. The multiclass SDP scales less favorably than the binary eigendecomposition.

AOC is promising where the observable itself has physical meaning: polarization/coherence analyzers [12], ambient-vibration covariance changes [13], order parameters in phase discovery [14], force/torque contact modes, and quantum device drift. Each domain requires its own nuisance model, instrument constraints, and real-data validation.

The decisive condition is not "machine learning" but a match between the state representation and the phenomenon: if the change lies in a low-rank operator moment while means cancel, a bounded quadratic witness can be both sample-efficient and interpretable. If the difference lies outside that moment, the method is blind by construction.

TABLE I
Selected Held-Out Results

Experiment/method	Metric	Mean
Ising linear logistic	accuracy	0.5000
Ising AOC	accuracy	0.9998
Ising RBF SVM	accuracy	0.9998
Structure mean logistic	ROC AUC	0.4985
Structure rank-one AOC	ROC AUC	0.9768
Structure oracle mode	ROC AUC	0.9774
Optics fixed H/V	success	0.5000
Optics Helstrom/Aer	success	0.9500

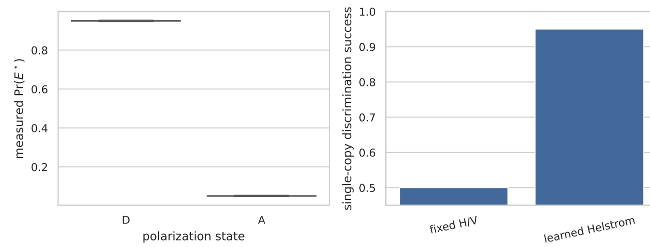

Fig. 3. Optical state discrimination. The optimal Bloch-axis analyzer replaces a deliberately blind horizontal/vertical analyzer; Aer sampling matches the analytic probability.

VI. Conclusion

AOC turns the qualitative goal of maximizing class difference into an exact measurement-design problem. The binary solution is a positive spectral projector, rank constraints select positive Ky Fan modes, multiclass classification is a POVM SDP, and all inputs can be accumulated and merged exactly. The experiments establish strong performance only in matched mean-blind physical regimes and expose ties and failures. The next step is to restrict observables to known symmetries or instrument algebras and resolve which physical sector carries the distinguishability.

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